DEVELOPMENT OF THYROID & PARATHYROID GLANDS AND RELATED ANOMALIES

LEARNING OBJECTIVES

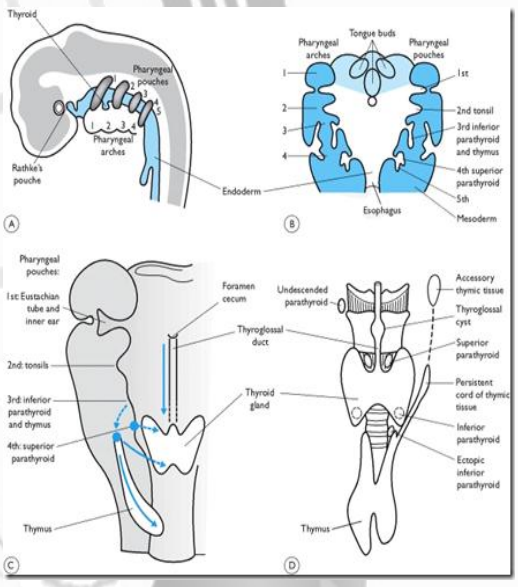
- At the end of lecture students should know:
  - Embryological development of thyroid and parathyroid glands
  - Related developmental anomalies

INITIAL EMBRYOLOGY

Thyroid:

- 1st endocrine gland
- Day ~24th gestation
- Proliferation of **endodermal** cells on midline pharyngeal floor
- Between 2 key structures:
  - tuberculum impar
  - Copula

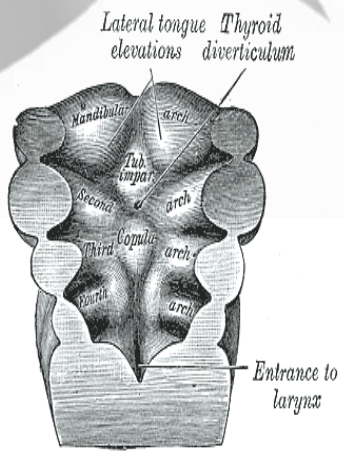
…called "**foramen cecum**"



INITIAL EMBRYOLOGY

Thyroid:

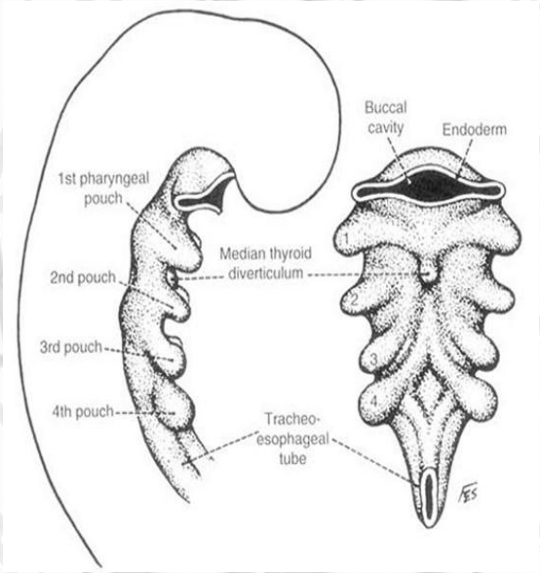
- Initially develops caudal to the **"tuberculum impar"** (median tongue bud)
- From the 1st pharyngeal arch
- Midline on floor of developing pharynx



INITIAL EMBRYOLOGY

- Foramen cecum begins ant to the copula, as a '**hypobranchial eminence**'
- Median embryologic swelling on mesoderm that arises from the **second** pharyngeal pouch (although the third and fourth pouches are also involved)

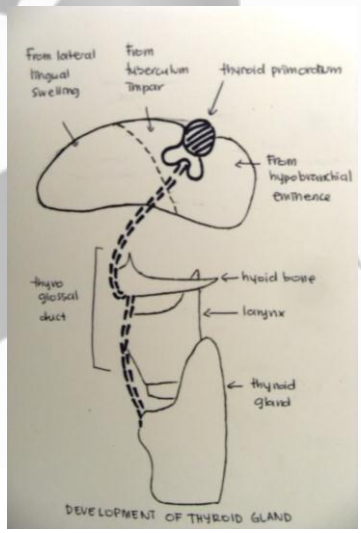
- Summary - thyroid gland originates from **b/w** 1st and 2nd pharyngeal pouches



INITIAL EMBRYOLOGY

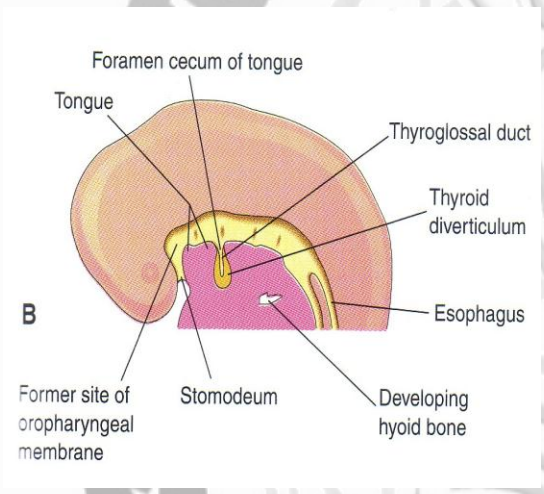
"***THYROID PRIMORDIUM***"

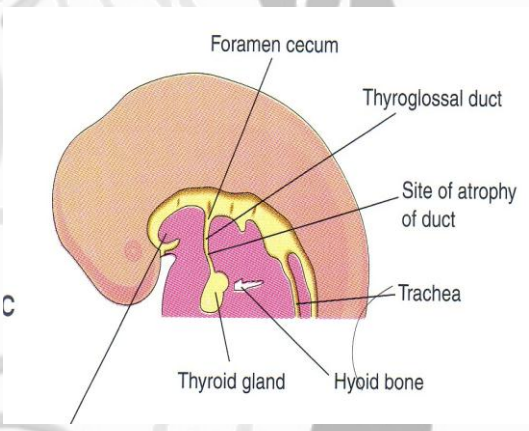
- initial thyroid precursor
- starts as a simple midline thickening → forms thyroid diverticulum
- initially hollow
- later solidifies and becomes bilobed



DEVELOPMENT OF THYROID GLAND

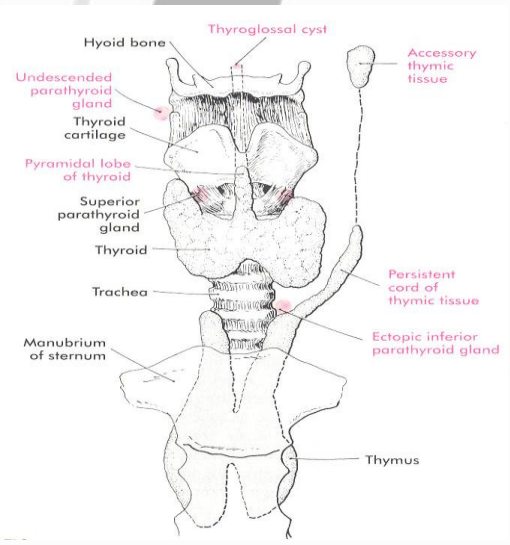
- Developed from endodermal cells of floor of pharynx (area of developing tongue).
- Development begin as diverticulum at dorsum of tongue between tuberculum impar and hypobranchial eminence.
- Downward elongation of thyroid diverticulum (form thyro-glossal duct) in front of hyoid and thyroid cartilage, ends by dividing into 2 parts.
- Ultimobranchial body gives rise to parafollicular cells of thyroid gland.





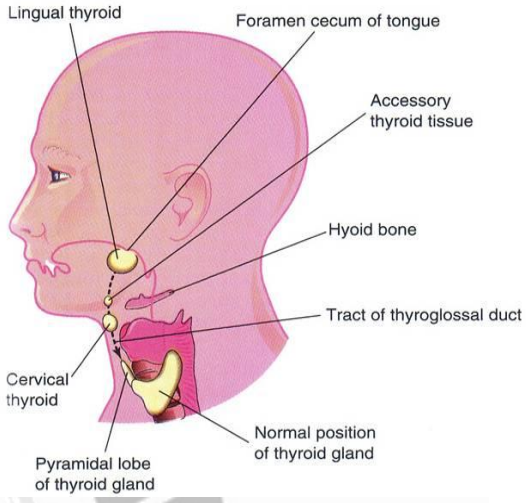
THYROID GLAND

- Around the 4th week to rest by the 7-8th week endoderm of the floor of mouth between the 1st and 2nd arches descends as a bilobed diverticulum from the foramen cecum



DESCENT OF THYROID GLAND

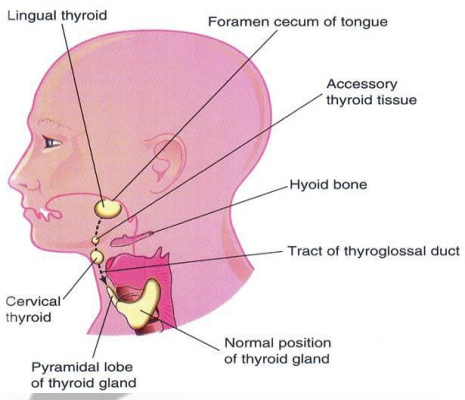
- Initial descent starts ant to pharyngeal gut
- Thyroid still connected to the tongue via the thyroglossal duct
- Later - tubular duct solidifies then obliterates entirely (during gestational weeks **7-10**)
- **Foramen cecum** = opening of the thyroglossal duct into the tongue



DESCENT OF THYROID GLAND

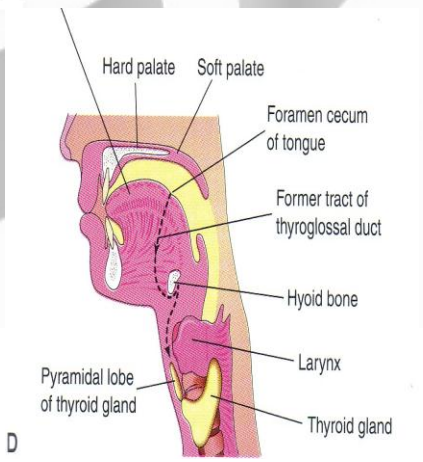
Pyramidal Lobe:

- in up to 50% people
- persistence of the inferior end of the thyroglossal duct (fix to obliterate)
- may be attached to the hyoid bone (thryoglossal duct cyst) OR
- may be incorporated *into* a thyroglossal duct cyst



DESCENT OF THYROID GLAND

- Further descent – ventral to the hyoid bone and laryngeal cartilage
- During descent – thyroid forms its **mature shape**: median isthmus connecting 2 lateral lobes
- Descent Complete at 7th gestational week



PARAFOLLICULAR CELLS

- Also known as **C cells**
- From **ultimobranchial body** (last structure derived from the branchial pouches)
- Ultimobranchial body – from **5th** pharyngeal pouch
- Secrete calcitonin (Ca regulation)

PARAFOLLICULAR CELLS

- Migrating cells from the neural crest region infiltrate to form ultimobranchial body (UBB)
- This structure (neural crest + UBB) = incorporated → thyroid gland
- UBB **fuses** with the thyroid gland and **disseminates** its cells into it

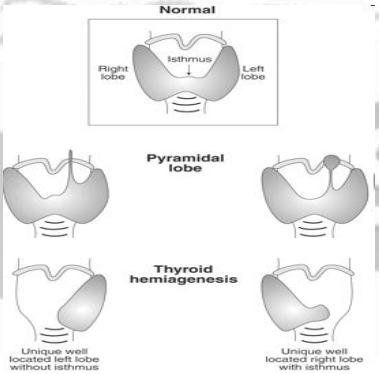
- ***Summary***: the C cells of the thyroid are of **neural crest** origin.

ANOMALIES OF THYROID GLAND

- ***ATHYROSIS*** or absence of the thyroid gland, is associated with ***cretinism*.**

- ***PYRAMIDAL LOBE*** most common developmental abnormality. The pyramidal lobe generally lies in the midline but can arise from either lobe. Origin from the left lobe is more common than the right lobe.

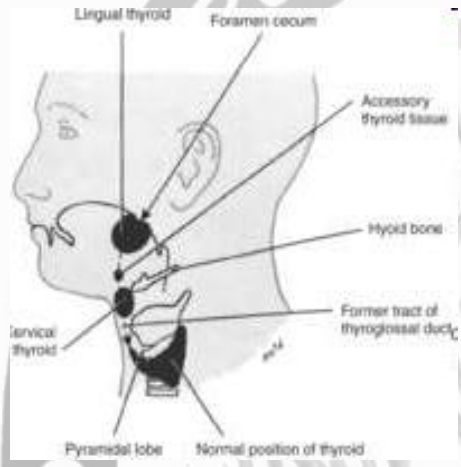
- ***THYROID HEMIAGENESIS***
- only one lobe of the thyroid is present .(Most commonly the left lobe)

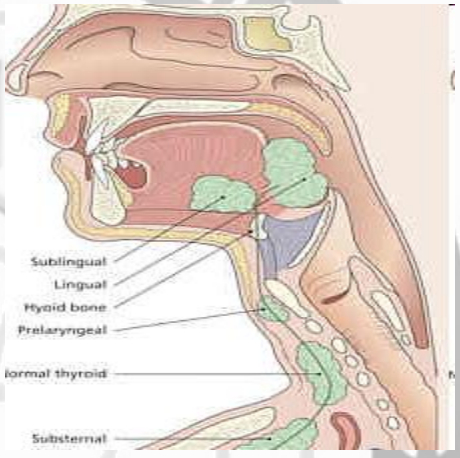


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ANOMALIES OF THYROID GLAND

4. ***Ectopic thyroid*** represent an arrest in the usual descent of part or all of the thyroid-forming material along the normal pathway.
   - most common at the base of the tongue '***lingual thyroid*** '



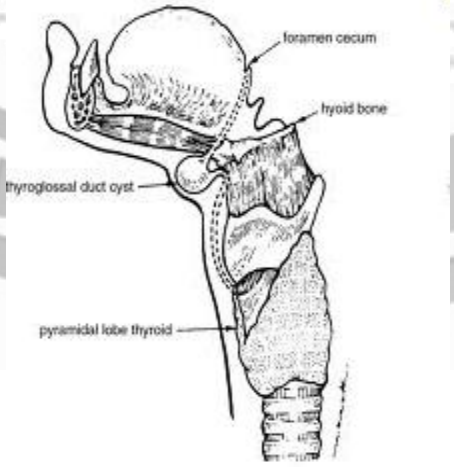


ANOMALIES OF THYROID GLAND

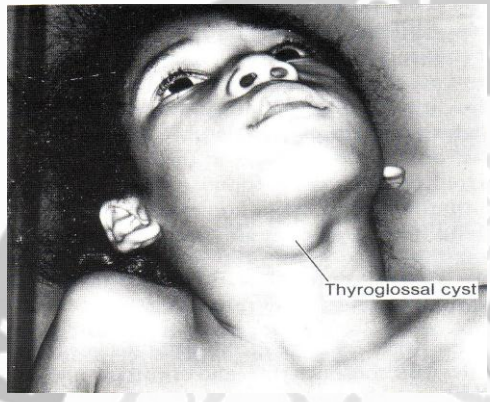
5.***THYROGLOSSAL DUCT CYST***

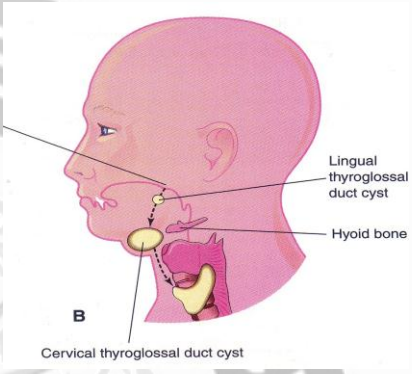
The thyroglossal duct occasionally persists as a cyst. Cysts usually appear in the midline or just off the midline between the isthmus of the thyroid and the hyoid bone.

- can form a sinus
- Rx – "Sistrunk" procedure (includes resection part of hyoid bone)



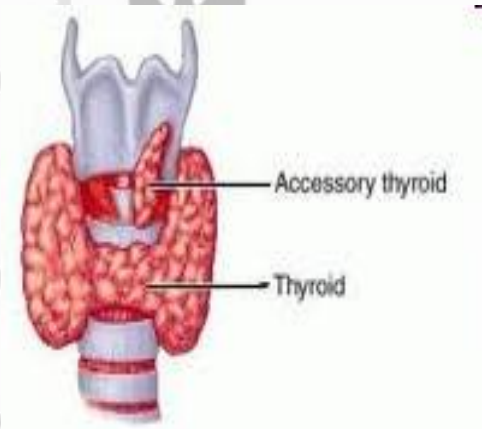
THYROGLOSSAL DUCT CYST





ANOMALIES OF THYROID GLAND

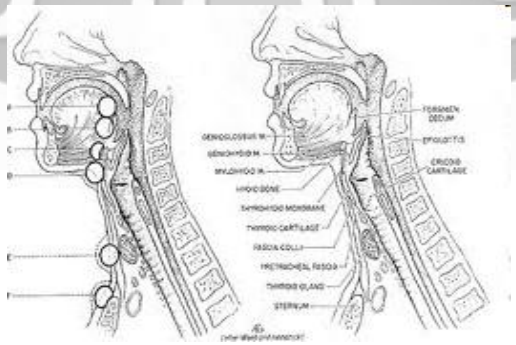
6. ***Accessory thyroid***
   - from remnants of thyroglossal duct
   - may be functional but insufficient for normal function of thyroid



7. ***Suprahyoid and Infrahyoid Thyroid***

   In these cases, thyroid tissue is present in a midline position above or below the hyoid bone.

   - **Hypothyroidism** with elevation of thyrotropin (TSH) secretion

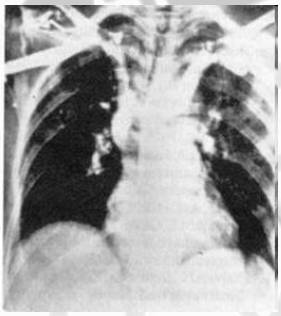


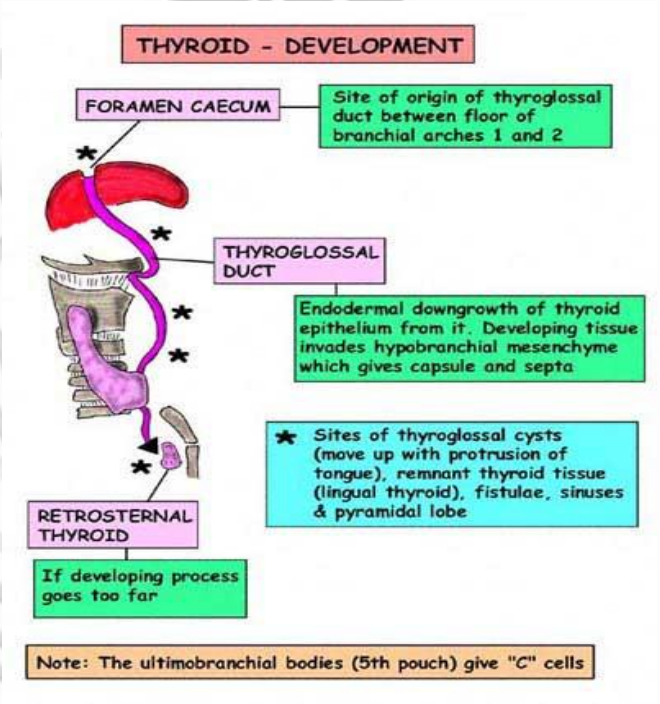
8. ***Substernal Goiters***

Developmental abnormalities may lead to the finding of thyroid tissue in the mediastinum or, rarely, even within the tracheal or esophageal wall.

- routine chest radiography
- may have dyspnea or dysphagia from tracheal or esophageal compression or displacement.







9. ***Aberrant thyroid (Retrosternal goiter)***

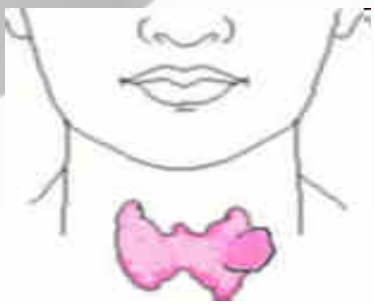
over-descend of thyroid gland in the thorax.

10. ***Thyroglossal fistula***

Thyroglossal duct opens to skin.

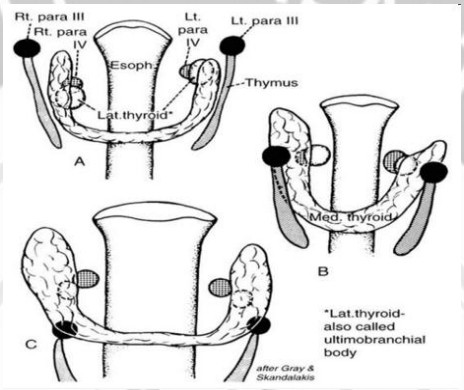
THYROID NODULES

- 4% of population
- 1/20 will harbor Cancer
- H&P combined with FNA is crucial for diagnosis
- FNA
  - Malignant
  - Suspicious
  - Benign
  - Indeterminat



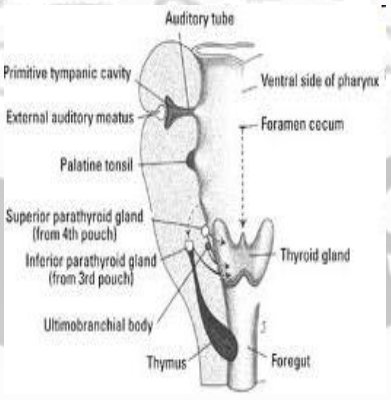
DEVELOPMENT OF PARATHYROID GLANDS

- The parathyroid glands derived from the third pouches descend with the thymus and are carried to a more inferior position than the parathyroid derived from the fourth pouches.
- This explains why the parathyroid glands derived from the third pair of pouches are located inferior to those from the fourth pouches



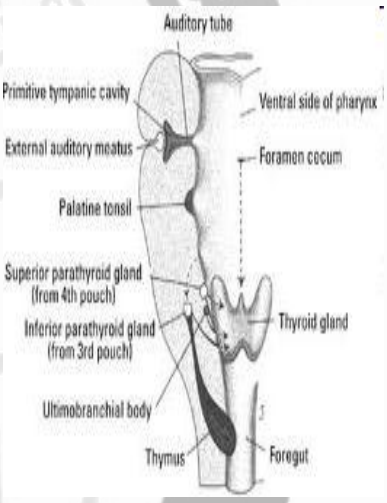
INFERIOR PARATHYROID EMBRYOLOGY

- arise from the dorsal wing of the third pharyngeal pouch.
- The third brachial arch differentiates, at gestational weeks 5-6, with the ventral wing becoming the thymus.
- The thymus and parathyroid both lose their connections to the pharynx at gestational week 7.
- The thymus then migrates caudally and medially, pulling the parathyroid with it; therefore, parathyroid IIIs are further inferior than are parathyroid IVs. The parathyroid in turn loses its connection with the thymus.
- The inferior parathyroid glands usually stop at the dorsal surface of the thyroid gland, outside of the fibrous capsule of the gland itself.



SUPERIOR PARATHYROID EMBRYOLOGY

- arise from the dorsal wing of the fourth pharyngeal pouch, differentiating at gestational weeks 5-6.
- At gestational week 7, the glands lose connections with the pharynx and attach themselves to the thyroid gland, which is migrating caudally, albeit far less a migration than the thymus (with parathyroid IIIs as described above).
- Because of the lesser length of migration, the superior parathyroid glands (IV) are in a more constant location than the inferior parathyroids (III).
- The superior parathyroids are generally located more posterior and medial than the inferior parathyroids, and their final resting point is usually on the dorsal surface of the thyroid gland, outside the fibrous capsule of the thyroid gland



HISTOGENESIS OF PARATHYROID GLANDS

- The epithelium of the dorsal parts of the third and fourth pouches proliferates during the fifth week
- Forms small nodules on the dorsal aspect of each pouch
- Vascular mesenchyme soon grows into these nodules, forming capillary network
- The chief or principal cells differentiate during the embryonic period and regulate fetal calcium metabolism
- The oxiphil cells differentiate 5 to 7 years after birth

HISTOGENESIS OF PARATHYROID GLANDS

- The elongated ventral part of each fourth pouch develops into ultimopharyngeal body

- Its cells disseminate within the thyroid gland, giving rise to parafollicular cells

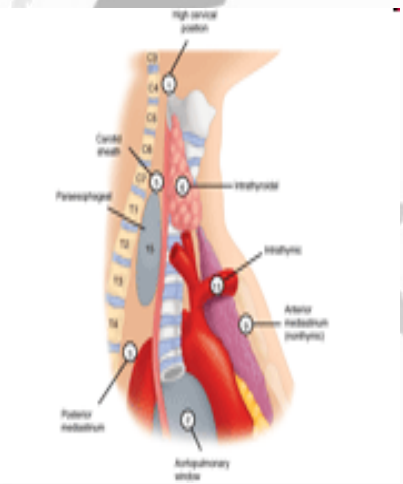
- They are also called “C” cells because they produce calcitonin that regulate normal calcium level in body fluids
- “C” cells differentiate from neural crest cells

DEVELOPMENTAL ANOMALIES

- ***Accessory or supernumerary parathyroid glands***
  - Approximately 13% of individuals at autopsy.
  - Result from tissue fragmentation during the migration of the glands.

- ***Absence of parathyroids*** (i.e <4 glands)
  - approximately 3% of individuals at autopsy.
  - result from a failure of the primordia to differentiate into the parathyroid glands or may be due to the parathyroid gland atrophy early in development.

- ***Ectopic parathyroid glands***
  - occur in 15-20% of patients.
  - located anywhere near or even within the thyroid or thymus.
  - **Common locations**
    - Bifurcation of the common carotid
    - Intrathorasic as low as the aortopulmonary window
    - Anterior mediastinum
    - Posterior mediastinum
    - Retroesophageal and
    - Prevertebral regions



- However, even when the parathyroid glands are in an ectopic location, they still often are symmetrical from side to side, making localization somewhat easier.

FATE OF THYRO-GLOSSAL DUCT

- Upper end of the duct remains as foramen cecum on dorsum of tongue.
- The duct degenerates from foramen cecum and hyoid bone.
- From the hyoid two divisions of the duct form levator glandulae thyroidae and may be pyramidal lobe.
- Divisions of the duct form lobes of thyroid gland with isthmus in between.
- Ultimobranchial body forms parafollicular cells of the gland.

DIGEORGE SYNDROME

- DiGeorge syndrome manifests as congenital thymic aplasia and absent parathyroid glands
- Results from failure of the third and fourth brachial pouches to differentiate.
- Also associated with facial abnormalities from abnormal development within first arch structures.
- No genetic cause is known, but teratogens are the assumed mechanism.
- *Symptoms* include neonatal tetany and impaired cellular immunity with normal humoral immunity



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